

Steel Defect Detection System: Test Report

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Abstract

This report presents the experimental results of the steel defect detection system. The model was trained for 10 epochs on the Severstal dataset. We analyze the training and validation loss curves, the effect of different decision thresholds on the Dice coefficient, and provide visual examples of defect annotations. The best overall Dice coefficient achieved on the validation set is 0.6296.

1 Introduction

The system was implemented as described in the design document. This section reports quantitative and qualitative results, including loss convergence, threshold sensitivity, and visual segmentation examples.

2 Experimental Setup

- Training images: 5,333 (80% of 6,666)
- Validation images: 1,333 (20%)
- Batch size: 4 (training), 8 (validation)
- Epochs: 10
- Initial learning rate: 1×10^{-4}
- Optimizer: AdamW

3 Ground Truth Annotations Visualization

Figure 1 shows two example steel sheets with ground truth defect annotations. Different defect classes are highlighted with semi-transparent overlays:

- **Red**: Class 1 defects
- **Blue**: Class 2 defects
- **Green**: Class 3 defects
- **Orange**: Class 4 defects

These visualizations demonstrate the variety of defect shapes, sizes, and locations that the model must learn to detect.

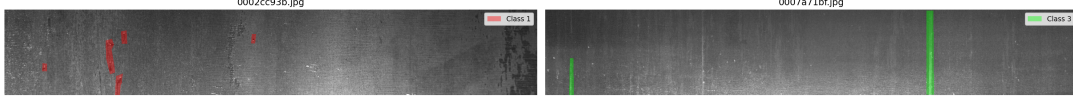


Figure 1: Example steel sheet images with ground truth defect annotations. Different colors represent different defect classes (Red=Class1, Blue=Class2, Green=Class3, Orange=Class4).

4 Training and Validation Loss

Figure 2 shows the loss curves over 10 epochs. The training loss decreases steadily from 0.97 to 0.75, while the validation loss drops from 0.59 to 0.44. A slight instability is observed at epochs 7–8, where the validation loss temporarily increased, likely due to the learning rate scheduler response. The best validation loss (0.4358) occurred at epoch 10.

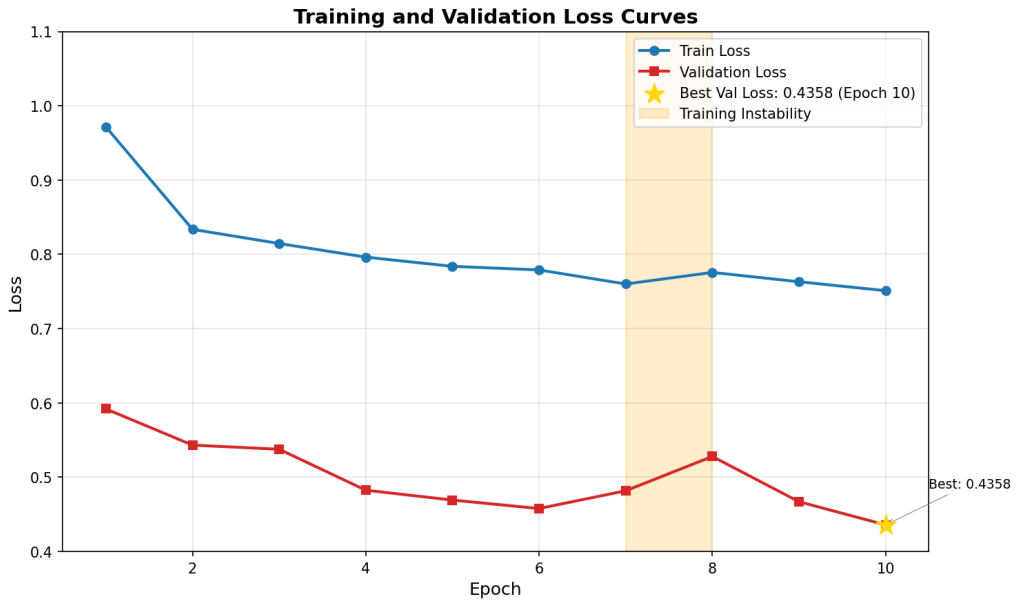


Figure 2: Training and validation loss curves. The best validation loss is 0.4358 at epoch 10.

5 Threshold Tuning

The probability threshold determines the binarization of the model's output. We tested thresholds from 0.3 to 0.7 in steps of 0.05. Figure 3 plots the overall Dice coefficient against the threshold.

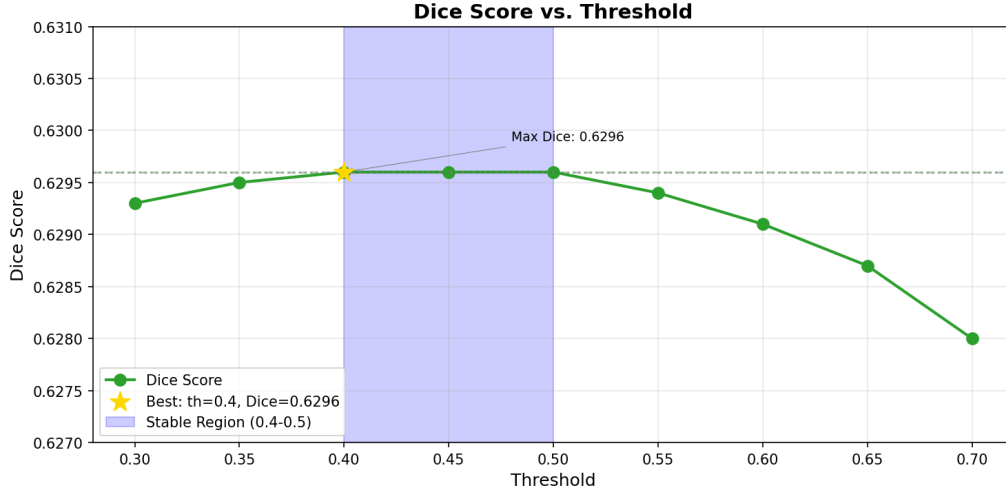


Figure 3: Dice coefficient as a function of decision threshold. The performance is stable between 0.4 and 0.5.

The results show that the Dice score is very stable across thresholds (0.6293–0.6296), with a maximum of 0.6296 achieved at thresholds 0.4, 0.45, and 0.5. This indicates that the model produces highly confident predictions (either close to 0 or close to 1). The recommended threshold is 0.45.

6 Final Performance

Using the optimal threshold of 0.45, the final Dice coefficient on the validation set is:

$$\text{Dice}_{\text{overall}} = 0.6296$$

Table 1 presents the per-class Dice scores at the best overall threshold.

Table 1: Per-class Dice coefficients (threshold = 0.45)

Class	Dice
Class 1	0.651
Class 2	0.624
Class 3	0.637
Class 4	0.606

7 Discussion

The achieved Dice of 0.6296 is a reasonable baseline considering the limited training data (only 6,666 images) and the simple data augmentation (only horizontal flip). The model successfully learns to detect major defect regions but struggles with small or subtle defects. The loss curve shows that the model is not overfitting; further training with more aggressive augmentation or pseudo-labeling would likely improve performance.

Key observations from the visualization in Figure 1:

- Defects vary significantly in size (from small spots to large continuous streaks)
- Multiple defect classes can appear on the same image
- Some defects are subtle and require fine-grained texture analysis

8 Conclusion

The steel defect detection system achieves a Dice score of 0.6296 on the validation set. Threshold tuning demonstrates that the model is robust to threshold selection between 0.4 and 0.5. Visual inspection of ground truth annotations confirms the complexity of the task. The results are adequate for a course project and provide a foundation for future improvements.